

Maternal smoking DNA methylation risk score associated with health outcomes in offspring of European and South Asian ancestry

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Abstract

Maternal smoking has been linked to adverse health outcomes in newborns but the extent to which it impacts newborn health has not been quantified through an aggregated cord blood DNA methylation (DNAm) score. Here we examine the feasibility of using cord blood DNAm scores leveraging large external studies as discovery samples to capture the epigenetic signature of maternal smoking and its influence on newborns in White European and South Asian populations. We first examined association between individual CpGs and cigarette smoking during pregnancy, smoking exposure in two White European birth cohorts ($n = 744$). Several previously reported genes for maternal smoking were supported, with the strongest and most consistent signal from the *GFI1* gene (6 CpGs with $p < 5 \times 10^{-5}$). Leveraging established CpGs for maternal smoking, we constructed a cord blood epigenetic score of maternal smoking that was internally validated in one of the European-origin cohorts ($n = 347$). This score was then tested for association with smoking status, secondary smoking exposure during pregnancy, and health outcomes in offspring measured after birth in an independent white European ($n = 397$) and a South Asian birth cohort ($n = 504$). The epigenetic maternal smoking score was strongly associated with smoking status during pregnancy ($OR=1.09$ [1.07,1.10], $p=1.96 \times 10^{-32}$) and more hours of self-reported smoking exposure per week (1.97 [1.22, 2.71], $p=2.80 \times 10^{-7}$) in White Europeans, but not with self-reported exposure ($p > 0.05$) in South Asians. The same score was consistently associated with

smaller birth size (-0.22 cm [-0.35, -0.083], $p=0.0016$) and lower birth weight (-0.05kg [-0.075, -0.025], $p=3.42\times 10^{-4}$) in the combined South Asian and White European cohorts. This cord blood epigenetic score can help identify babies exposed to maternal smoking and assess its long-term impact on growth. Notably, these results indicate a consistent association between the DNAm signature of maternal smoking and a small body size and low birthweight in newborns, in both white European mothers who exhibited some amount of smoking and in South Asian mothers who themselves were not active smokers.

eLife assessment

This study offers a **useful** advance by introducing a cord blood DNA methylation score for maternal smoking effects, with the inclusion of cohorts from diverse backgrounds. However, the overall strength of evidence is deemed **incomplete**, due to concerns regarding low exposure levels and low statistical power, which hampers the generalisability of their findings. The study provides an interesting basis for future studies, but would benefit from the addition of more cohorts to validate the findings and a focus on more diverse health outcomes.

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Introduction

Maternal smoking has adverse effects on offspring health including pre-term delivery (1,2), stillbirth (3), and low birth weight (4), and is associated with pregnancy complications such as maternal higher blood pressure, and gestational diabetes (5). Consistent with the Developmental Origins of Health and Disease (DOHaD) hypothesis, maternal smoking exposes the developing fetus to harmful chemicals in tobacco that negatively impact the health of newborns, resulting in early-onset metabolic diseases, such as childhood obesity (6–9). Yet self-reported smoking status is subject to underreporting among pregnant women (10–12). This could subsequently impact the effectiveness of interventions aimed at reducing smoking during pregnancy and may skew data on the risks associated with maternal smoking.

DNA methylation is one of the most commonly studied epigenetic mechanisms by which cells regulate gene expression, and is increasingly recognized for its potential as a biomarker (13). Differential DNA methylation has been established as a reliable biochemical response to cigarette smoking and was shown to capture the long-lasting effects of persistent smoking in ex-smokers (14–16). Recent large epigenome-wide association studies (EWAS) have robustly identified differentially methylated cytosine-phosphate-guanine (CpG) sites associated with adult smoking (15,17,18) and maternal smoking (19,20). Our recent systematic review of 17 cord blood epigenome-wide association studies (EWAS) found that out of the 290 CpG sites reported to be associated with at least one of the following: maternal diabetes, pre-pregnancy body mass index (BMI), diet during pregnancy, smoking, and gestational age, 19 sites were identified in more than one study and all of them associated with maternal smoking (21). Furthermore, these findings have led to a more thorough investigation of the epigenetic mechanisms underlying associations between well-established epidemiological exposures and outcomes, such as the relationship between maternal smoking and birth weight in Europeans (20,22–25) and the less studied African American populations (24) as well as between maternal diet and cardiovascular health (26).

Only a handful of cohort studies were designed to assess the influence of maternal exposures on DNA methylation changes in non-Europeans (24,27). It has been suggested that systematic patterns of methylation (28), such as cell composition, could differ between individuals of different ancestral backgrounds, which could in turn confound the association between differential DNAm and smoking behaviours (29). These systematic differences also contribute to different smoking-related methylation signals at individual CpGs (30). Thus, a comparative study of maternal smoking exposure is a first step towards generalizing existing EWAS results to other populations and a necessary step towards addressing health disparities that exist between populations due to societal privilege, including race or ethnicity and socioeconomic factors.

A promising direction in epigenetic studies of adult smoking is the application of a methylation score (31); this strategy can also be applied to disseminate current knowledge on differential DNA methylation studies of maternal smoking. A methylation score is usually tissue-specific and combines information from multiple CpGs using statistical models (13). Reducing the number of predictors and measurement noise in the data can lead to better statistical power and a more parsimonious instrument for subsequent analyses. It is also of interest to determine whether methylation scores demonstrate the capacity to predict outcomes in diverse human populations, given the presence of systematic differences in methylation patterns due to ancestral backgrounds (28).

In this paper, we investigated the epigenetic signature of maternal smoking on cord blood DNA methylation in newborns, as well as its association with newborn and later life outcomes in one South Asian which refers to people who originate from the Indian subcontinent, and two predominantly European-origin birth cohorts. Similar to the Born in Bradford study (32), we observed several differentiating epidemiological characteristics between South Asian and European-origin mothers. Notably, almost none of the South Asian mothers were current smokers and had low smoking rates pre-pregnancy as compared to European mothers, which is consistent with the broader trends of lower smoking rates in South Asian females (33). Another relevant observation is the small birth size and low birth weight in the South Asian newborns. These differences in newborn size and weight may be influenced by various factors, including maternal nutrition, genetics, and socioeconomic status. Keeping these differences in mind, we first conducted cohort-specific epigenetic association studies between available CpGs and maternal smoking in the predominantly European-origin cohorts, benchmarking with previously identified CpGs for maternal smoking and adult smoking. Second, we leveraged the reported summary statistics from existing large EWASs to construct a methylation risk score (MRS) for maternal smoking. The MRS was first internally validated in one of the European-origin cohorts and then tested in a second independent European-origin cohort. Third, we examined the association between maternal smoking MRS and newborn health outcomes, including length, weight, BMI ponderal index, and early-life anthropometrics in both European and South Asian cohorts.

Materials and Methods

Study population

The NutriGen Alliance is a consortium consisting of four prospective, population-based birth cohorts that enrolled birthing mother and newborn pairs in Canada. Details of these cohorts have been described elsewhere (34). The current investigation focused on i). European-origin offspring from the population-based CHILD study who were selected for methylation analysis, ii). The Family Atherosclerosis Monitoring In early life (FAMILY) study that is predominately European-origin, and iii). The SouTh Asian biRth cohort (START) study that is exclusively comprised of people who originated from the Indian subcontinent known as South Asians. The ethnicity of the parents was self-reported and recorded at baseline in all three cohorts. Biological samples, clinical assessments, and questionnaires were used to derive health phenotypes and an

array of genetic, epigenetic, and metabolomic data. The superordinate goal of the NutriGen study is to understand how nutrition, environmental exposures, and physical health of mothers impact the health and early development of their offspring using a multi-omics approach.

Methylation data processing and quality controls

Newborn cord blood samples were processed using two methylation array technologies. About half of the START samples and selected samples from CHILD were hybridized to the Illumina Human-Methylation450K BeadChip (HM450K) array, which covers CpGs in the entire genome (35). The raw methylation data were generated by the Illumina iScan software and separately pre-processed for START and CHILD using the “sesame” R package following pipelines designed for HM450K BeadChip (36). The FAMILY samples were profiled using a targeted array based on the Infinium Methylation EPIC designed by the Genetic and Molecular Epidemiology Laboratory (GMEL; Hamilton, Canada). The GMEL customized array includes ~3000 CpG sites that were previously reported to associate with complex traits or exposures and was designed to maximize discovery while keeping the costs of profiling epigenome-wide DNA methylation down. The targeted methylation data were pre-processed using a customized quality control pipeline and functions from the “sesame” R package recommended for EPIC.

Pre-processed data were then used to derive the β -value matrix, where each column gives the methylation level at a CpG site as a ratio of the probe intensity to the overall probe intensity. Additional quality control filters were applied to the final beta-value matrices to remove samples with > 10% missing probes and CpG probes with >10% samples missing. Cross-reactive probes and SNP probes were removed as recommended for HM450 (37) and EPIC arrays (38,39). For CpG probes with a missing rate <10%, mean imputation was used to fill in the missing values. We further excluded samples that were either mismatches between reported sex and methylation-inferred sex or were duplicates. Finally, considering the low prevalence of smokers, we sought to reduce spurious associations by removing non-informative probes that were either all hypomethylated (β -value < 0.1) or hypermethylated (β -value > 0.9), which have been shown to have less optimal performance (40). A summary of the sample and probe inclusion/exclusion is shown in Supplementary Table 1. A detailed description of pre-processing and quality control steps is included in Supplementary Material.

Cell-type proportions (CD8T, CD4T, Natural Killer cells, B cells, monocytes, granulocytes, and nucleated red blood cells) were estimated following a reference-based approach developed for cord blood (41) and using R packages “*FlowSorted.CordBloodCombined.450k*” and “*FlowSorted.Blood.EPIC*”. All data processing and subsequent analyses were conducted in R v.4.1.0 (42).

Phenotype data processing and quality controls

At the time of enrollment, all pregnant women completed a comprehensive questionnaire that collected information on prenatal diet, smoking, education, socioeconomic factors, physical activities and health as detailed previously (43,44). Maternal smoking history (0=never smoked, 1=quit before this pregnancy, 2=quit during this pregnancy, or 3=current smoker) was assessed during the second trimester (at baseline). Smoke exposure was measured as “number of hours exposed per week”. GDM was determined based on a combination of oral glucose tolerance test (OGTT), self-report, and reported diabetic treatments (insulin, pills, or restricted diet). For South Asian mothers in START, the same OGTT threshold as Born in Bradford (27,32) was used, while the International Association of the Diabetes and Pregnancy Study Groups (IASDPSG) criteria (45) for OGTT were used in CHILD and FAMILY cohorts. Mode of delivery (emergency c-section vs. other) was collected at the time of delivery.

Newborn length and weight were collected immediately after birth and extracted from medical chart. The newborns were then followed up at 1, 2, 3, and 5 years of age and provided basic anthropometric measurements, including height, weight, hip and waist circumference, BMI, sum of the skinfolds (triceps skinfold and subscapular skinfold). Additional phenotypes included smoking exposures (hours per week) at home, potential allergy based on mother reporting any of: eczema, hay fever, wheeze, asthma, food allergy (egg, cow milk, soy, other) for her child in FAMILY and START, and asthma based on mother's opinion in CHILD ("In your opinion, does the child have any of the following? Asthma").

Phenotype and Methylation Data Consolidation

The current investigation examines the impact of maternal smoking or smoke exposure on DNA methylation derived from newborn cord blood in START and the two predominately European cohorts (CHILD and FAMILY). To maximize sample size in FAMILY and CHILD, we retained either self-identified or genetically confirmed Europeans based on available genetic data (Supplementary Table 1). The cohorts consist of representative population samples without enrichment for any clinical conditions, though only singleton mothers were invited to participate. For continuous phenotypes, an analysis of variance (ANOVA) using the F-statistics or a two-sample *t*-test was used to compare the mean difference across the three cohorts or two groups, respectively. For categorical phenotypes, a chi-square test of independence was used to compare the difference in frequencies of observed categories. Note that three of the categories under smoking history in the START cohort had expected cell counts less than 5, and was thus excluded from the comparison, the reported *p*-value was for CHILD and FAMILY.

The final analytical datasets, after combining the quality-controlled methylation data and phenotypic data, included 352, 411, and 504 mother-newborn pairs from CHILD, FAMILY, and START, respectively. Demographic characteristics and relevant covariates of the epigenetic subsample and the overall sample are summarized in **Table 1** and Supplementary Table 2, respectively.

Epigenome-Wide Association of Maternal Smoking in European Cohorts

Since there were no current smokers in START (**Table 1**), we tested the association between maternal smoking and differential methylated sites in FAMILY (# of CpG = 2,544) and CHILD (# of CpG = 200,050). The primary outcome variable was "current smoker", defined by mothers self-identified as currently smoking during the pregnancy vs. those who never smoked or quit either before or during pregnancy. We also included a secondary outcome variable "ever smoker", defined by mothers who are current smokers or have quit smoking vs. those who never smoked. A tertiary outcome was smoking exposure, measured by the number of hours per week reported by the expectant mothers, and was available in all cohorts. We summarized the type of analyses for different outcomes in Supplementary Table 3.

We first conducted a separate epigenetic association study in each cohort, testing the association between methylation β -values at individual CpGs and the smoking phenotype using either a logistic regression model for smoking status or a linear regression for smoking exposure as the outcome. The model adjusted for additional covariates including the estimated cord blood cell proportions, maternal age, social disadvantage index, which is a continuous composite measure of social and economic exposures (46), mother's years of education, GDM, and parity. The smoking exposure variable was skewed, and a rank-based transformation was applied to mimic a standard normal distribution.

Table 1.

Characteristics of the epigenetic subsample (1,267 mother-newborn pairs) from the CHILD, FAMILY, START cohorts.

	Phenotypes	CHILD (N=352)	FAMILY (N=411)	START (N=504)	ANOVA F-test or Chi-squared test P-value for differences
Mother	Smoking History				
	never smoked	247 (70.2%)	253 (61.6%)	501 (99.4%)	<0.001*
	quit before this pregnancy	72 (20.5%)	58 (14.1%)	1 (0.2%)	
	quit during this pregnancy	17 (4.8%)	57 (13.9%)	1 (0.2%)	
	currently smoking	11 (3.1%)	29 (7.1%)	0 (0%)	
	Missing	5 (1.4%)	14 (3.4%)	1 (0.2%)	
	Smoking Exposure (hr/week)				
	Mean (SD)	0.97 (± 7.64)	2.52 (± 12.83)	0.33 (± 2.67)	<0.001
	Missing	12 (3.4%)	5 (1.2%)	42 (8.3%)	
	Gestational Diabetes Mellitus				
	YES	16 (4.5%)	66 (16.1%)	183 (36.3%)	<0.001
	NO	336 (95.5%)	345 (83.9%)	320 (63.5%)	
	Missing	0 (0%)	0 (0%)	1 (0.2%)	
	Years of Education				<0.001
	Mean (SD)	16.96 (± 3.08)	16.85 (± 3.39)	15.81 (± 2.41)	
	Missing	7 (2.0%)	3 (0.7%)	0 (0%)	
	Mother's Age				
	Mean (SD)	32.69 (± 4.45)	31.86 (± 5.42)	30.12 (± 3.91)	<0.001
	Missing	4 (1.1%)	0 (0%)	0 (0%)	
	Parity				
	Mean (SD)	0.72 (± 0.88)	0.80 (± 1.02)	0.80 (± 0.81)	0.0
	Missing	2 (0.6%)	0 (0%)	13 (2.6%)	
	Pre-pregnancy BMI (kg/m²)				
	Mean (SD)	24.78 (± 5.42)	26.46 (± 6.38)	23.71 (± 4.45)	<0.001
	Missing	132 (37.5%)	16 (3.9%)	2 (0.4%)	
	Newborn Sex				
	Male	194 (55.1%)	211 (51.3%)	239 (47.4%)	0.0
	Female	158 (44.9%)	200 (48.7%)	265 (52.6%)	
	Plant Based Diet				
	Mean (SD)	-0.48 (± 0.46)	0.19 (± 0.67)	1.56 (± 1.14)	<0.001
	Missing	23 (6.5%)	36 (8.8%)	16 (3.2%)	
	Health Conscious Diet				
	Mean (SD)	0.21 (± 0.81)	-0.73 (± 0.73)	-0.42 (± 0.79)	<0.001
	Missing	23 (6.5%)	36 (8.8%)	16 (3.2%)	
	Western Diet				
	Mean (SD)	-0.15 (± 0.63)	1.06 (± 1.20)	-0.51 (± 0.65)	<0.001
	Missing	23 (6.5%)	36 (8.8%)	16 (3.2%)	
Newborn	Gestational Age (weeks)				
	Mean (SD)	39.53 (± 1.38)	39.44 (± 1.47)	39.20 (± 1.32)	<0.001
	Missing	4 (1.1%)	0 (0%)	0 (0%)	
	Birth Length (cm)				
	Mean (SD)	51.68 (± 2.52)	50.20 (± 2.16)	51.44 (± 2.69)	<0.001
	Missing	71 (20.2%)	10 (2.4%)	7 (1.4%)	
	Birth Weight (kg)				
	Mean (SD)	3.50 (± 0.49)	3.53 (± 0.50)	3.26 (± 0.46)	<0.001
	Missing	6 (1.7%)	0 (0%)	1 (0.2%)	
	Newborn BMI (kg/m²)				
	Mean (SD)	13.11 (± 1.41)	13.94 (± 1.29)	12.31 (± 1.39)	<0.001
	Missing	72 (20.5%)	10 (2.4%)	7 (1.4%)	
	Newborn Ponderal Index (kg/m³)				

	Mean (SD)	25.45 (± 3.14)	27.79 (± 2.55)	24.02 (± 3.17)	<0.001
	Missing	72 (20.5%)	10 (2.4%)	7 (1.4%)	
Estimated cell proportions	CD8T				
	Mean (SD)	0.01 (± 0.01)	0.04 (± 0.03)	0.02 (± 0.02)	<0.001
	CD4T				
	Mean (SD)	0.11 (± 0.06)	0.13 (± 0.06)	0.16 (± 0.07)	<0.001
	NK				
	Mean (SD)	0.02 (± 0.02)	0.03 (± 0.03)	0.02 (± 0.03)	<0.001
	Bcell				
	Mean (SD)	0.02 (± 0.02)	0.04 (± 0.03)	0.04 (± 0.03)	<0.001
	Mono				
	Mean (SD)	0.01 (± 0.02)	0.04 (± 0.03)	0.03 (± 0.03)	<0.001
	Gran				
	Mean (SD)	0.80 (± 0.10)	0.60 (± 0.13)	0.72 (± 0.14)	<0.001
	nRBC				
	Mean (SD)	0.08 (± 0.08)	0.12 (± 0.11)	0.07 (± 0.11)	<0.001
	MNLR				
	Mean (SD)	6.59 (± 6.00)	3.30 (± 3.14)	3.98 (± 3.08)	<0.001
	Missing	6 (1.7%)	0 (0%)	3 (0.6%)	

* comparison for CHILD and FAMILY only

Table 1. (continued)

We then meta-analyzed association results for maternal smoking status in the European cohorts using an inverse variance-weighted fixed-effect model. The meta-analysis was conducted for 2,112 CpGs that were available in both CHILD (profiled using HM450K) and FAMILY (profiled using the targeted array). For the tertiary outcome, we conducted an inverse variance meta-analysis including START using both a fixed-effect model. For each EWAS or meta-analysis, the false discovery rate (FDR) adjustment was used to control multiple testing and we considered CpGs that passed an FDR-adjusted p -value < 0.05 to be relevant for maternal smoking.

Using DNA Methylation to Construct Predictive Models for Maternal Smoking

We sought to construct a predictive model in the form of a methylation risk score (MRS) using reported associations of maternal smoking. The proposed solution adapted the existing lassosum method (47) that was originally designed for polygenic risk scores, where the matrix of SNP genotypes (X) can be conveniently replaced by the β -value matrix. For more details, see the Supplementary Material. Briefly, an objective function under elastic-net constraint was minimized to obtain the elastic-net solution y , where only summary statistics (b) and a scalar of the covariance between the β -values of the CpGs ($X^T X$) are needed. The tuning parameters A and A_2 were chosen by validating on the observed smoking history (as a continuous outcome) in CHILD that produced the most significant model. The optimized A_1 and A_2 were then used to create a final model that entails a list of CpGs and their corresponding weights, which were then used to calculate a MRS for maternal smoking in the FAMILY and START samples.

The summary statistics of the discovery EWAS were obtained from EWAS catalog (<http://www.ewascatalog.org/>) reported under “PubMed ID 27040690” by Joubert and colleagues (19). The summary statistics were restricted to the analysis “sustained maternal smoking in pregnancy effect on newborns adjusted for cell composition”. Of the 2620 maternal smoking CpGs that passed the initial screening, 2,107 were available in CHILD but only 128 were common to CHILD, FAMILY, and START. To evaluate whether the targeted GMEL-EPIC array design has comparable performance as the epigenome-wide array to evaluate the epigenetic signature of maternal smoking, a total of three MRSs were constructed, two using the 128 CpGs available in all cohorts – across the HM450K and targeted GMEL-EPIC arrays – and with either CHILD ($n = 347$ with non-missing smoking history) or FAMILY ($n = 397$ with non-missing smoking history) as the validation cohort, and another using 2,107 CpGs that were only available in CHILD and START samples with CHILD as the validation cohort. The validation model considered the continuous smoking history without modification as the outcome, while accounting for covariates, which included the estimated cord blood cell proportions, maternal age, social disadvantage index, mother’s years of education, GDM, and parity. Henceforth, we referred to these derived maternal smoking scores as the FAMILY targeted MRS, CHILD targeted MRS, and the HM450K MRS, respectively. To benchmark and compare with existing maternal smoking MRSs, we calculated the Reese score using 28 CpGs (48, 49), Richmond score using 568 CpGs (49), Rauschert score using 204 CpGs (50), Joubert score using all 2,620 CpGs with evidence of association for maternal smoking (19), and finally a three-CpG score for air pollution (51). The details of these scores and score weight can be found in Supplementary Table 4.

Statistical analysis

For each cohort, we contrasted the three versions of the derived scores using an analysis of variance analysis (ANOVA) along with pairwise comparisons using a two-sample t -test to examine how much information might be lost due to the exclusion of more than 10-fold CpGs at the validation stage, in all samples and in non-smokers. We also examined the correlation structure between all derived and external MRSs using a heatmap summarizing their pairwise Pearson’s correlation coefficient. Then, we compared the mean difference of each MRS score among smoking history using an ANOVA F -test and two-sample t -test to understand whether there was a dosage dependence in the cord blood DNAm signature of maternal smoking. Additionally, each

score was tested against a binary outcome for current smoker vs. not, and two continuous measures for smoking history and weekly smoking exposure. The binary outcome was tested using a logistic regression model and the predictive performance was assessed using area under the receiver operating characteristic curve (AUC). The continuous outcome was examined using a linear regression model and its performance was quantified using the adjusted R^2 . For the derived MRS, we empirically assessed whether the resulting score was portable to the START cohort by examining the distribution of the final score among the overall samples and never smokers. The comparison examined a mean difference using an analysis of variance F -test, a variance difference using a Levene's test, and the overall distribution using a non-parametric Anderson-Darling test. Finally, we tested the association between each maternal smoking MRS and smoking phenotypes in mothers, as well as offspring phenotypes using a linear regression model, when applicable, adjusting for the child's age at each visit. The association results were meta-analyzed for phenotypes with homogeneous effects across the cohorts using a fixed-effect model.

Results

Cohort Sample Characteristics

The analyses included 763 European mother-child pairs with cord blood DNAm data from the CHILD study (CHILD; $n = 352$)[\(52\)](#) and The Family Atherosclerosis Monitoring In earLY life (FAMILY; $n = 411$) study [\(43\)](#), and 503 South Asian mother-child pairs from The SouTh Asian biRth cohOrT (START) study [\(44\)](#). We observed lower past smoking and missingness on smoking history among pregnant women in START as compared to CHILD or FAMILY using the epigenetic subsample (**Table 1**) and the overall sample (Supplementary Table 2). Pregnant women in START were significantly different from CHILD or FAMILY in that they were on average younger at delivery, had a lower BMI, and a higher rate of GDM, in line with other cohort studies in South Asian populations [\(53,54\)](#). As compared to START, newborn infants from CHILD and FAMILY had a longer gestational period, a higher birth weight, and a higher BMI at birth (**Table 1**; Supplementary Table 2). We observed no difference between cohorts in terms of parity or newborn sex in the epigenetic subsample (**Table 1**).

Within the European epigenetic subsample, of the 744 mother-newborn pairs with complete smoking history data, 40 (5.3%) newborns were exposed to current maternal smoking, which is on the lower end of the spectrum for the prevalence of smoking during pregnancy (9.2%-32.5%) among Canadians [\(55\)](#). In addition, mothers who smoked during pregnancy were on average younger, had fewer years of education, and had higher household exposure to smoking (Supplementary Table 6). However, there was no statistically significant difference between newborns exposed to current and none or previous smoking in terms of birth weight, birth length, gestational age, or estimated cord blood cell proportions.

Epigenetic Association of Maternal Smoking in White Europeans

The two predominantly White European cohorts, FAMILY ($n = 397$) and CHILD ($n = 347$), contributed to the meta-analysis of maternal smoking for both the primary outcome of current smoking (**Figure 2-A**) and the secondary outcome of ever smoking (Supplementary Figure 1). The top associated CpGs with current maternal smoking were mapped to the growth factor independent 1 (*GFI1*) gene on chromosome 1, with cg12876356 as the lead (meta-analyzed effect = -1.11 ± 0.22 ; meta-analyzed $p = 2.6 \times 10^{-6}$; FDR adjusted $p = 0.006$; **Table 2**). There were no CpGs associated with the ever-smoker status at an FDR of 0.05, though the top signal (cg09935388) was also mapped to the *GFI1* gene (Pearson's r^2 correlation with cg12876356 = 0.75 and 0.68 in CHILD and FAMILY, respectively; Supplementary Figure 1). The top associated CpG from the meta-analysis of smoking exposure (hours per week) in the European-origin cohorts (**Figure 2-B**) was cpg01798813 on chromosome 17, which was also associated with maternal smoking and was consistent in the direction of association (meta-analyzed effect = -0.18 ± 0.04 ; meta-analyzed $p =$

1.4×10^{-5} ; FDR adjusted $p = 0.04$; **Table 2** [↗](#)). There was no noticeable inflation of empirical type I error in the association p-values from the meta-analysis, with the median of the observed association test statistic roughly equal to the expected median (Supplementary Figure 2). As a sensitivity analysis, we repeated the analysis for the continuous smoking exposure under rank transformation vs. raw phenotype for the associated CpG in *GFI1* and examined the regression diagnostics (Supplementary Material), and found that the model under rank-transformation deviated less from assumptions. Further, we observed consistency in the direction of association for the 128 CpGs that overlapped between our meta-analysis and the 2,620 CpGs with evidence of association for maternal smoking (19 [↗](#)) (Supplementary Figure 3). Specifically, the Pearson's correlation coefficient for maternal smoking and weekly smoking exposure was 0.72 and 0.60, respectively. The maternal smoking and smoking exposure EWASs in CHILd alone did not yield any CpGs after FDR correction (Supplementary Figure 4).

Methylation Risk Score (MRS) Captures

Maternal Smoking and Smoking Exposure

The final MRSs, validated using CHILd European samples ($n = 347$), included 15 and 143 CpG markers (Supplementary Table 7) from the targeted array and the epigenome-wide HM450 array, respectively. Both produced methylation scores that were significantly associated with maternal smoking history (ANOVA F-test p -values $= 1.0 \times 10^{-6}$ and 2.4×10^{-14} in CHILd and 6.9×10^{-16} and $< 2.2 \times 10^{-16}$ in FAMILY; **Figure 3** [↗](#)), and the best among alternative scores for CHILd and FAMILY (Supplementary Table 5). With the exception of the air pollution MRS, all remaining scores were marginally associated with smoking history in both CHILd and FAMILY (Supplementary Figure 5) and correlated with each other (Supplementary Figure 6). There was no statistically significant difference in mean between the two scores in any of the three cohorts (two-sample t-test $ps > 0.6$) or among non-smokers (two-sample t-test $ps > 0.6$; Supplementary Figure 7). Since the HM450 score provides statistically more significant results in both CHILd and FAMILY with smoking history, despite the reduction in CpGs included (only 10 out of 143 CpGs present in FAMILY; Supplementary Table 5), we proceeded with the HM450 MRS model constructed using the 143 CpGs in subsequent analyses.

The HM450 MRS was significantly associated with maternal smoking history in CHILd and FAMILY ($n = 397$), but not in START ($n = 503$; **Figure 3** [↗](#)) – not surprisingly – due to the low number of ever-smokers ($n = 2$). A weak dose-dependent relationship between the MRS and the four categories of maternal smoking status in the severity of exposure ([0] = never smoked; [1] = quit before this pregnancy; [2] = quit during this pregnancy; [3] = currently smoking) was present in CHILd but was not replicated in FAMILY (**Figure 3** [↗](#)). The areas under the Receiver Operating Characteristic (ROC) curve (area under the curve or AUC) for detecting current smokers were 0.95 and 0.89 in CHILd and FAMILY, respectively, while the AUCs for detecting ever-smokers were 0.61 and 0.60 (Supplementary Table 5), respectively. Meanwhile, the maternal smoking MRS was significantly associated with increased number of hours exposed to smoking per week in the two White European cohorts (1.97 ± 0.38 hours per 1 unit of increase in MRS, FDR adjusted $p = 4.2 \times 10^{-6}$; Supplementary Table 8; cohort specific $p = 0.0087$ in CHILd and 0.51 ± 0.15 , $p = 6.9 \times 10^{-5}$ in FAMILY; **Table 3** [↗](#)), but not in the South Asian birth cohort ($p = 0.41$; Supplementary Table 8).

Among individuals who had never smoked, no statistically significant mean difference was observed in the distribution of the combined methylation score between South Asian and European cohorts (Supplementary Table 9). These results provided empirical support for the portability of an European-derived maternal smoking methylation score to South Asian populations.

Figure 1.

Schematic overview of the analytical pipeline for the cord blood DNAm maternal smoking score and association study.

* indicates cohort sample size including those with missing smoking history.

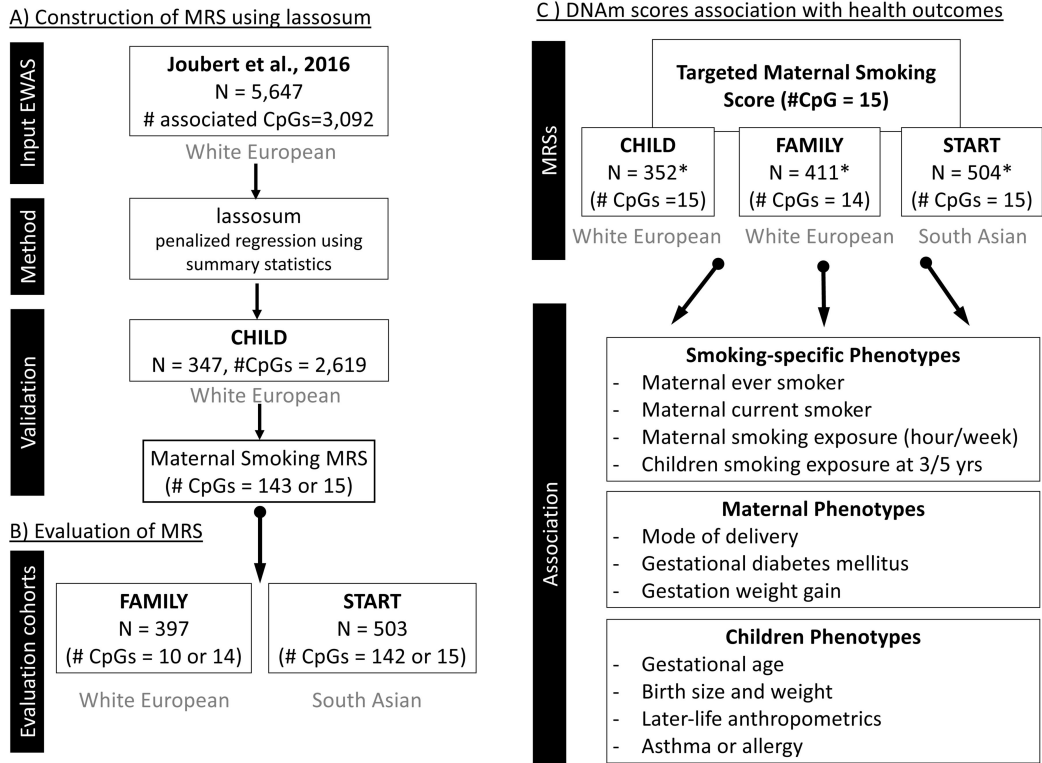
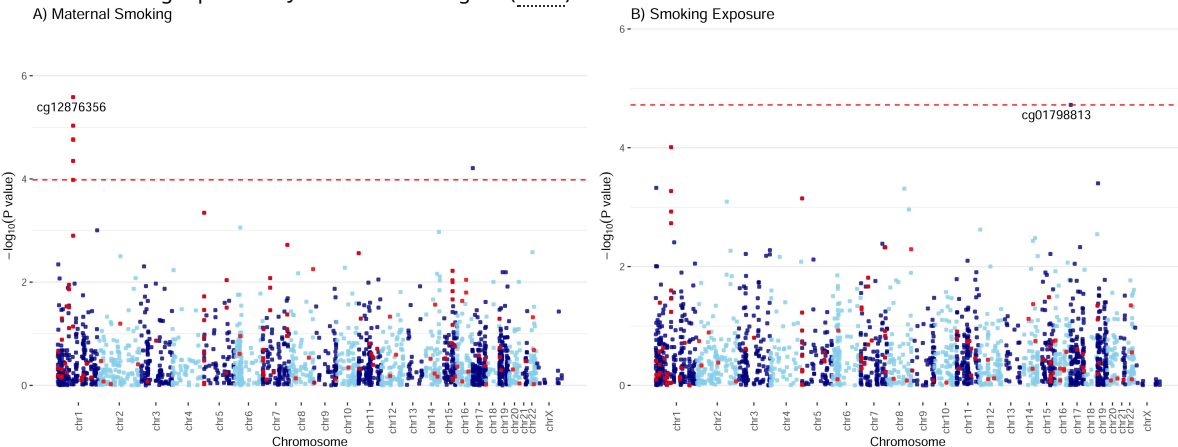


Figure 2.

Manhattan plots of the meta-analyzed association between cord blood DNAm and maternal smoking in Europeans.

Manhattan plots summarized the meta-analyzed association p -values between cord blood DNA methylation levels and current maternal smoking (A) or smoking exposure (B) at a common set of 2,114 CpG sites. The red line denotes the smallest $-\log_{10}(p\text{-value})$ that is below the FDR correction threshold of 0.05. The red dots represent established associations with maternal smoking reported in Joubert and colleagues (19).



	CHR	Position	CpG	UCSC Reference Gene	Meta-analysis (CHILD and FAMILY)					Cohort-specific association P-value		Reported Association EWAS catalog
					Fixed Effect	Standard Error	Association P-value	P-value for Effect Heterogeneity	FDR adjusted Association P-value	CHILD	FAMILY	
Maternal Smoking	1	92481269	cg12876356	<i>GFI1</i>	-1.11	0.22	7.33E-07	0.51	0.0019	0.02	9.45E-06	MS;S; AC; BW
	1	92482032	cg09935388	<i>GFI1</i>	-1.15	0.24	2.26E-06	0.52	0.0029	0.02	2.71E-05	MS;GA; S; AC; BMI; BW
	1	92482405	cg14179389	<i>GFI1</i>	-1.48	0.32	5.03E-06	0.73	0.0035	0.01	1.12E-04	MS;S
	1	92481144	cg18146737	<i>GFI1</i>	-0.92	0.20	5.58E-06	0.50	0.0035	0.04	3.95E-05	MS;S; AC; BW
	1	92480576	cg09662411	<i>GFI1</i>	-0.94	0.22	1.64E-05	0.29	0.0083	0.10	3.85E-05	MS;S
	1	92481479	cg18316974	<i>GFI1</i>	-0.74	0.18	3.58E-05	0.33	0.0152	0.13	7.34E-05	MS;S; AC; BW
	17	2494783943	cg01798813	–	-0.83	0.21	1.09E-04	0.34	0.0395	0.02	0.0016	A; GA; BMI
Smoking Exposure	1	92482032	cg09935388	<i>GFI1</i>	-0.18	0.04	1.39E-05	0.23	0.04	0.15	2.45E-05	MS;GA; S; AC; BMI; BW
	17	2494783943	cg01798813	–	-0.18	0.04	3.30E-05	0.13	0.04	0.00035	0.013	A; GA; BMI

MS: maternal smoking

GA: gestational age

AC: alcohol consumption

BMI: body mass index

T2D: type 2 diabetes

A: age

BW: birth weight

Table 2.

Meta-analysis results of association between CpGs and maternal smoking and smoking exposure that passed a marginal $p < 0.05$ threshold after the false discovery rate correction in European cohorts.

Figure 3.

Relationships between maternal smoking MRS and maternal smoking history categories for each of the studies.

Maternal smoking methylation score (y-axis) was shown as a function of maternal smoking history (x-axis) in levels of severity for prenatal exposure for each study. Each severity level was compared to the never-smoking group and the corresponding two-sample *t*-test *p*-value was reported. The analysis of variance via an F-test *p*-value was used to indicate whether a mean difference in methylation score was present among all smoking history categories. The sample size for START cohort was provided due to the low counts in categories of any smoking.

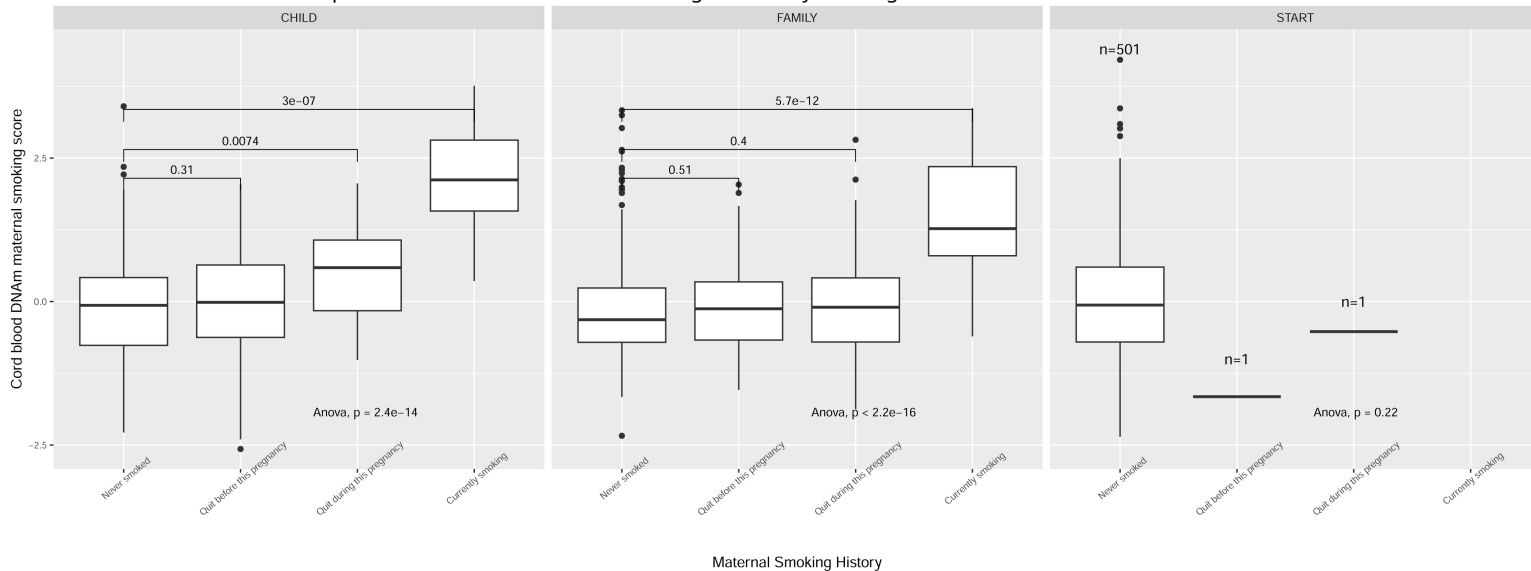


Table 3.

Significant associations between maternal smoking methylation risk score and phenotypes in CHILD, FAMILY and START.

	Meta-analysis (CHILD, FAMILY, START)					Cohort-specific association P-value		
	Fixed Effect	Standard Error	Association P-value	FDR adjusted Association P-value	P-value for Effect Heterogeneity	CHILD	FAMILY	START
Smoking Exposure (hr/week)	0.29	0.12	0.012	0.032	7.32E-07	5.40E-05	3.41E-05	0.58
Gestational Age (weeks)	0.30	0.12	0.011	0.032	1.70E-07	6.32E-05	6.38E-06	0.59
Birth Length (cm)	-0.22	0.07	0.0016	0.0076	0.33	0.35	0.16	0.0023
Birth Weight (kg)	-0.05	0.01	3.42E-04	0.0021	0.66	0.016	0.026	0.094
1-year Height (cm)	-0.38	0.09	2.00E-05	5.99E-04	0.89	0.047	0.0070	0.0079
1-year Weight (kg)	-0.09	0.04	0.018	0.044	1.00	0.16	0.17	0.19
2-year Height (cm)	-0.39	0.13	0.0018	0.0076	0.50	0.72	0.059	0.0067
3-year Skinfold thickness	0.52	0.14	1.08E-04	0.0015	0.24	0.014	0.0011	0.38
5-year Height (cm)	-0.43	0.17	0.0090	0.030	0.94	0.16	0.07	0.21
5-year Skinfold thickness	0.50	0.18	0.0061	0.023	0.59	0.019	0.089	0.77

Association between MRS and other phenotypes

The maternal smoking MRS was consistently associated with increasing weekly smoking exposure in children reported by mothers at the 1-year visit (0.44 ± 0.15 , $p = 0.0044$) in CHILD, and at 3-year visit (0.86 ± 0.26 , $p = 0.0037$; Supplementary Table 8) in FAMILY, but not in START as everyone reported non-exposure to smoking. A higher maternal smoking MRS was significantly associated with lower birth weight (-0.05 ± 0.01 , $p = 3.4 \times 10^{-4}$) and smaller birth size (-0.22 ± 0.07 , $p = 0.0016$) in the combined European and South Asian cohorts (Table 3; Supplementary Table 8). The meta-analysis revealed no heterogeneity in the direction nor the effect size of associations for body size and weight between populations at birth or at later visits (heterogeneity p -values = 0.18–1; Supplementary Table 8). The association between the MRS and several children phenotypes, including height or length, weight, and skinfolds, appeared to persist with similar estimated effects throughout early developmental years (Supplementary Table 8), albeit the most significant effects were at birth, and the significance attenuated at later visits. We did not find any association with self-reported allergy or asthma in children at later visits (Supplementary Table 8). Further, there was no evidence of association between the MRS and any maternal outcomes (Supplementary Table 8).

Discussion

We examined the epigenetic signature of maternal smoking and smoking exposure using newborn cord blood samples from predominately European-origin and South Asian cohorts via two strategies: an individual CpG-level EWAS approach, and a multivariate approach in the form of a methylation score. The EWAS results replicated the association between maternal smoking and CpGs in the *GFI1* gene that is well described in the literature with respect to smoking (15, 17), maternal smoking (19, 20, 56, 57), and birth weight (23). In the latter case, we observed a significant association with maternal smoking history and smoking exposure in European-origin newborns. Further, we noted a weak dose-dependent relationship between maternal smoking history and the methylation score in one European cohort (CHILD) but this was not replicated in the other (FAMILY). Since the timing and duration of maternal smoking during pregnancy were not directly available, these differences could play a role in the magnitude and specificity of DNA methylation changes in cord blood. Finally, the significant association of the MRS with the newborn health metrics in START, in the absence of mothers' active smoking, could be the result of underreporting of smoking, poor recall of the time of quitting, and/or due to air pollution exposure (58), leading to oxidative stress. This suggests that our cord blood DNAm signature of maternal smoking is perhaps not unique to cigarette smoking, but captures similar biochemical responses, for example, via the aryl hydrocarbon receptor (59, 60). Our observation that a higher MRS was associated with lower birth weight and smaller birth length in both ethnic populations is thus consistent with the established link between oxidative stress and metabolic syndrome (61).

Contrary to DNA methylation studies of smoking in adults, where whole blood is often used as a proxy tissue, there are multiple relevant tissues for maternal smoking during pregnancy, including the placenta of the mother, newborn cord blood, and children's whole blood. However, methylation changes measured in whole blood or placenta of the mother, or cord blood of infants showed substantially different patterns of association signals (62). There are several advantages of using a cord blood based biomarker from the DoHaD perspective. Firstly, cord blood provides a direct reflection of the *in utero* environment and fetal exposure to maternal smoking. Additionally, since cord blood is collected at birth, it eliminates potential confounding factors such as postnatal exposures that may affect maternal blood samples. Furthermore, studying cord blood DNAm allows for the assessment of epigenetic changes specifically relevant to the newborn, offering valuable information on the potential long-term health implications. Meanwhile, methylation signals are known to be tissue-specific, thus it would be of interest for future research to combine

differential methylation patterns from all relevant tissue to assess the immediate and long-term effects of maternal smoking. Another direction to further this line of research is to explore postnatal factors that mitigate prenatal exposures, for example, breastfeeding, which has been shown to have a protective effect against maternal tobacco smoking (63 [↗](#)). Indeed, more research is necessary to understand the critical periods of exposure and the dose-response relationship between maternal smoking and cord blood DNA methylation changes. Ongoing efforts to monitor the offspring and collect data in the next decade are in progress to establish the long-term association between maternal smoking and cardio-metabolic health (43 [↗](#), 44 [↗](#)). As such, the constructed MRS can facilitate future research in child health and will be included as part of the generated data for others to access.

The strengths of this report include ethnic diversity, and fine phenotyping in a prospective and harmonized way with follow-up at multiple early childhood stages. This work is the first major multi-ancestry study that utilizes methylation scores to study maternal smoking and examines their portability from European-origin populations to South Asians. The use of MRS, as compared to individual CpGs, is a powerful tool to systematically investigate the influence of DNA methylation changes and whether it has lasting functional consequences on health outcomes. Our results converge with previous findings that epigenetic associations of maternal smoking are associated with newborn health, and add to the small body of evidence that these relationships extend to non-European populations and that different ancestral populations can experience the early developmental periods differently.

A few limitations should be mentioned. In the context of existing epigenetic studies of maternal smoking, we were not able to replicate signals in other well-reported genes such as *AHRR*, *CYP1A1*, and *MYO1G*, however, the MRS was able to pick up signals from these genes (Supplementary Table 7). This could be due to several reasons. First, the customized array with a limited number of CpGs (<3,000) was designed in 2016 and many large EWASs on smoking and maternal smoking conducted more recently had not been included. Nonetheless, we have shown that from a multivariate perspective, the MRS constructed using a targeted approach that was carefully designed can be equally powerful with the advantage of being cost-effective. Second, contrary to existing EWASs where the methylation values are typically treated as the outcome, and the exposure, such as smoking, as the predictor; we reversed the regression such that the methylation levels were the predictors and smoking exposure as the outcome. This reverse regression approach is robust and our choice to reverse the regression was motivated by the goal of constructing a smoking score that combines the additive effects at multiple CpGs, which would otherwise be unfeasible. Third, systematic ancestral differences in DNA methylation patterns had been shown to vary at individual CpGs in terms of their association with smoking (30 [↗](#)). Converging with this conclusion, we also found the association with *GFI1* to be most consistent after adjusting for cell composition. Finally, maternal smoking is often associated with other confounding factors, such as socioeconomic status, other lifestyle behaviours, and environmental exposures. While we have done our best to control for well-known confounders that were available by study design, as in all observational studies, we could not account for unknown confounding effects. Finally, in recent years, maternal smoking is on a decline as a result of changes in social norms and public health policies (64 [↗](#)). This is also consistent with the lower smoking rates observed in our European cohorts (CHILD and FAMILY).

In conclusion, the epigenetic maternal smoking score we constructed was strongly associated with smoking status during pregnancy and self-reported smoking exposure in White Europeans, and with smaller birth size and lower birth weight in the combined South Asian and White European cohorts. The proposed cord blood epigenetic signature of maternal smoking has the potential to identify newborns who were exposed to maternal smoking *in utero* and to assess the long-term impact of smoking exposure on offspring health. In South Asian mothers with minimal smoking

behaviour, the relationship between the methylation score and negative health outcomes in newborns is still apparent, indicating that DNA methylation response is sensitive to smoking exposure, even in the absence of active smoking.

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Data availability statement

The summary statistics used to construct methylation risk scores are available from EWAS catalog at <http://www.ewascatalog.org/?trait=maternal%20smoking%20in%20pregnancy> with additional filters of PubMedID 27040690 and analysis on “Sustained maternal smoking in pregnancy effect on newborns adjusted for cell composition”.

Summary statistics generated in the current study, including a total of 8 primary association studies (three smoking phenotypes in three cohorts) and 3 sets of meta-analyzed results in Europeans are available upon request. All scripts to reproduce and validate the predictive model can be found at <https://github.com/WeiAkaneDeng/EpigeneticResearch/tree/main/MaternalSmoking>.

Conflicts of interest

No conflict of interest.

Ethics Statement

Ethical approval was obtained independently from the Hamilton Integrated Research Ethics Board: CHILD (REB 07–2929), FAMILY (REB 02–060), and START (REB 10–640). CHILD was additionally approved by the respective Human Research Ethics Boards at McMaster University, the Universities of Manitoba, Alberta, and British Columbia, and the Hospital for Sick Children. Legal guardians of each participant provided written informed consent. Written informed consent was obtained from the parent/guardian (participating mother) for each study separately. We also have now obtained additional ethics board approval from HiREB (REB 16592) for using the data from the three cohorts together without additional consent from the participants.

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Reviewer #2 (Public Review):

Summary:

The authors generated a DNA methylation score in cord blood for detecting exposure to cigarette smoke during pregnancy. They then asked if it could be used to predict height, weight, BMI, adiposity and WHR throughout early childhood.

Strengths:

The study included two cohorts of European ancestry and one of South Asian ancestry.

Weaknesses:

- (1) Numbers of mothers who self-reported any smoking was very low likely resulting in underpowered analyses.
- (2) Although it was likely that some mothers were exposed to second-hand smoke and/or pollution, data on this was not available.
- (3) One of the European cohorts and half of the South Asian cohort had DNA methylation measured on only 2500 CpG sites including only 125 sites previously linked to prenatal smoking.

<https://doi.org/10.7554/eLife.93260.2.sa1>

Reviewer #3 (Public Review):

Summary:

Deng et al. assess neonatal cord blood methylation profiles and the association with (self-reported) maternal smoking in multiple populations, including two European (CHILD, FAMILY) and one South Asian (START), via two approaches: 1) they perform an independent epigenome-wide association study (EWAS) and meta-analysis across the CHILD and FAMILY cohort, during which they also benchmark previously reported maternal-smoking associated sites, and 2) they generate new composite methylation risk scores for maternal smoking, and assess their performance and association with phenotypic characteristics in the three populations, in addition to previously described maternal smoking methylation risk scores.

Strengths and weaknesses:

Their meta-analysis across multiple cohorts and comparison with previous findings represents a strength. In particular the inclusion of a South Asian birth cohort is commendable as it may help to bolster generalizability. However, their conclusions are limited by several important weaknesses:

- (1) the low number of (self-reported) maternal smokers in particular their South Asian population, resulting in an inability to conduct benchmarking of maternal smoking sites in this cohort. As such, the inclusion of the START cohort in certain figures is not warranted (e.g., Figure 3) and the overall statement that smoking-associated MRS are portable across populations are not fully supported;
- (2) different methylation profiling tools were used: START and CHILD methylation profiles were generated using the more comprehensive 450K array while the FAMILY cohort blood samples were profiled using a targeted array covering only 3,000, as opposed to 450,000 sites, resulting in different coverage of certain sites which affects downstream analyses and MRS, and importantly, omission of potentially relevant sites as the array was designed in 2016 and substantial additional work into epigenetic traits has been conducted since then;
- (3) the authors train methylation risk scores (MRS) in CHILD or FAMILY populations based on sites that are associated with maternal smoking in both cohorts and internally validate them in the other cohort, respectively. As START cohort due to insufficient numbers of self-reported

maternal smokers, the authors cannot fully independently validate their MRS, thus limiting the strength of their results.

Overall strength of evidence and conclusions:

Despite these limitations, the study overall does explore the feasibility of using neonatal cord blood for the assessment of maternal smoking. However, their conclusion on generalizability of the maternal smoking risk score is currently not supported by their data as they were not able to validate their score in a sufficiently large number of maternal smokers and never smokers of South Asian populations.

While their generalizability remains limited due to small sample numbers and previous studies with methylation risk scores exist, their findings may nonetheless provide the basis for future work into prenatal exposures which will be of interest to the research community. In particular their finding that the maternal smoking-associated MRS was associated with small birth sizes and weights across birth cohorts, including the South Asian birth cohort that had very few self-reported smokers, is interesting and the authors suggest these findings could be associated with factors other than smoking alone (e.g., pollution), which warrant further investigation and would be highly novel. Future exploration should also include a strong focus on more diverse health outcomes, including respiratory conditions that may have long-lasting health consequences.

<https://doi.org/10.7554/eLife.93260.2.sa0>

Author response:

The following is the authors' response to the original reviews.

We thank the reviewers for their time and the thoughtful reviews on our manuscript. The reviewers brought good points regarding the sample size, and the low exposure in the South Asian cohort owing to their unique cultural and social practices. We recognize these as limitations of the paper and discussed these in the revised version. In the revised manuscript, we have taken the key suggestions by reviewers to 1) better illustrate the analytical flow and statistical methods, in particular, to show which datasets had been used in discovery, validation, and testing of the score – as a main figure in the manuscript and in the graphical abstract; 2) demonstrate there is no possibility of overfitting in our approach using statistical metrics of performance; 3) emphasize the goal was not for discovery (e.g. our own EWAS was not used for deriving the score), but to compare with existing EWASs and contrast the results from the white European and SA populations; 4) and supplement the analysis with previously derived maternal smoking, smoking and air pollution methylation score and to explore additional health outcomes in relation to lung health in newborns. Finally, we would also like to take this opportunity to re-iterate that it was not our objective to derive the most powerful methylation score of smoking nor to demonstrate the causal role of maternal smoking on birth weight via DNAm. We have restructured the manuscript as well as the discussion to clarify this. Please find below a point-by-point response to the comments below.

Reviewer #1:

The manuscript could benefit from a more detailed description of methods, especially those used to derive MRS for maternal smoking, which appears to involve overfitting. In particular, the addition of a flow chart would be very helpful to guide the reader through the data and analyses. The FDR correction in the EWAS corresponds to a fairly liberal p-value threshold.

We thank the reviewer for these good suggestions. In the revised manuscript, we have provided a flow chart as the new Figure 1, more detailed description of the method (added a subsection “Statistical analysis” under Materials and Methods) as well as metrics including measures of fit indices such as AUC and adjusted R² for each validation and testing dataset to illustrate there is no danger of overfitting (in new Supplementary Table 5).

The choice of use FDR was indeed arbitrary as there has been no consensus on what significance threshold, if any, should be used in the context of EWAS. Here we simply followed the convention in previous studies to contrast the top associated signals for their effects between different populations and with reported effect sizes. Throughout the manuscript, we have removed the notion of significant associations and used the phrase “top associated signals” or “top associations” when discussion EWAS results for individual CpGs.

Reviewer #2:

(1) The number of mothers who self-reported any smoking was very low, much lower than in the general population and practically non-existent in the South Asian population. As a result, all analyses appeared to have been underpowered. It is possibly for this reason that the authors chose to generate their DNA methylation model using previously published summary statistics. The resulting score is not of great value in itself due to the low-powered dataset used to estimate covariance between CpG sites. In fact, a score was generated for a much larger, better-powered dataset several years ago (Reese, EHP, 2017, PMID 27323799).

We thank the reviewer for pointing out the low exposure in the South Asian population, which we believe is complementary to the literature on maternal smoking that almost exclusively focused on white Europeans. However, the score was validating in the white European cohort (CHILDS; current smoking 3.1%), which was reasonably similar to the trend that maternal cigarettes smoking is on the decline from 2016 to 2021, from 7.2% to 4.6% (Martin, Osterman, & Driscoll, 2023). This is also consistent with the fact that CHILDS participants were recruited from major metropolises of Canada with relatively high SES and education as compared to FAMILY.

We do agree with the reviewers that a higher prevalence of maternal smoking in the validating sample could potential improve the power of the score. Our original analytical pipeline focused on CHILDS as the validation dataset; FAMILY (see the new Figure 1) was used as the testing data. We alternatively provided an analytical scheme using FAMILY as the validation dataset, as it had a higher proportion of current smokers, however, this is limited by the number of CpGs available (128 in FAMILY vs. 2,619 in CHILDS out of the 2,620 CpGs from (Joubert et al., 2016)). The results of all possible combinations of validation vs. testing and restriction of targeted array vs. HM450 are summarized in the new new Supplementary Table 5 and Supplementary Figure 5.

To clarify, our choice to construct DNAm score using published summary statistics was not an ad-hoc decision due to the observed low power from CHILDS EWAS. We agree with the reviewer that our study was indeed underpowered and was not originally intended for EWAS discovery. Thus, we specifically proposed to adopt a multivariate strategy from the literature of polygenic risk scores. This approach enabled us to leverage well-powered association signals without individual-level access to data with a sample size of $n > 5,000$ (Joubert et al., 2016). In comparison, the Reese maternal smoking score (Reese et al., 2017) had a discovery sample size of only $n = 1,057$. Our score was not out-performed, in fact, the AUC in both FAMILY (external validating dataset; $n=411$) and CHILDS (external testing dataset; $n=352$) and was larger than that based on the Reese score as tabulated below (part of the new Supplementary Table 5).

Author response table 1.

		Targeted score CHILD	HM450K score CHILD	Reese	
Discovery EWAS	Source	Joubert et al., 2016	Joubert et al., 2016	Reese et al., 2017	
	Tissue	Cord blood	Cord blood	Cord blood	
	Sample Size	5,647 newborns	5,647 newborns	1,057 newborns	
	Phenotype	Maternal smoking	Maternal smoking	Sustained maternal smoking	
	# of CpGs included	128	2,619	195	
Validation Data	Source	CHILD	CHILD	Reese et al., 2017	
	Method	elastic-net	elastic-net	Lasso	
	Sample Size	347	347	221 newborns	
	# of CpGs in score	15	143	28	
	Adjusted R2 with continuous smoking history	7.2%	11.4%	–	
	Adjusted R2 with smoking exposure	1.5%	4.3%	–	
	AUC for ever smoker status	0.62	0.61	–	
	AUC for current smoker status	0.89	0.95	0.9	
	Source	FAMILY	FAMILY	CHILD	FAMILY
	# of CpGs in score	14	10	24	7
Testing Data	Adjusted R2 with continuous smoking history	9.1%	9.1%	6.9%	0.3%
	Adjusted R2 with smoking exposure	4.3%	4.6%	3.4%	0.8%
	AUC for ever smoker status	0.61	0.6	0.59	0.50
	AUC for current smoker status	0.88	0.89	0.94	0.72

Further, regarding the comment on the covariance matrix. Indeed, lassosum via elastic-net and summary data requires a reference covariance matrix that is consistent between the discovery data and external validation data. In fact, for moderately sized correlation/covariance values ($r^2 > 0.1$), a sample size of >100 is sufficiently powered to detect it being different from 0 and thus used for estimation. Similar to the linkage disequilibrium of genotype data, the CpGs also exhibit a block-wise correlation structure and thus the theoretical framework of lassosum extends naturally to MRS.

In the revised manuscript, we included the Reese score, as well as a few additional scores to compare their predictiveness of smoking phenotypes in white European cohorts. We note that the applicability was limited in the FAMILY cohort that was profiled using a targeted array and only 7 out of 28 of the CpGs in the Reese score were available. As a result, though the Reese score had similar performance than our derived score in CHILD (0.94 vs. 0.95), its performance in FAMILY was compromised (0.72 vs. 0.89).

(2) The conclusion that "even minimal smoking exposure in South Asian mothers who were not active smokers showed a DNAm signature of small body size and low birthweight in newborns" is not warranted because no analyses were performed to show that the association between DNA methylation and birth size/weight was driven by maternal smoking.

We thank the reviewer for this subtle point – it was not our intention to suggest there was a causal relationship between DNA methylation and birth size that was mediated by maternal

smoking. We meant to suggest that the maternal smoking methylation score was consistently associated with negative outcomes in newborns of both white European and South Asian mothers despite no maternal smoking was present in South Asian mothers. It is possible that maternal smoking MRS was capturing a lot more than just smoking and second-hand smoking, such as other environmental exposures that also lead to oxidative stress. These together are associated with reduced birth size/weight.

In the revised manuscript, we have modified the conclusion above to:

“Notably, these results indicate a consistent association between the DNAm signature of maternal smoking and a small body size and low birthweight in newborns, in both white European mothers who exhibited some amount of smoking and in South Asian mothers who themselves were not active smokers.”

(3) Although it was likely that some mothers were exposed to second-hand smoke and/or pollution, data on this was either non-existent or not included in this study. Including this would have allowed a more novel investigation of the effects of smoke exposure on the pregnancies of non-smoking mothers.

We agree with this comment – second-hand smoking was captured by self-reported weekly smoking exposure by the mothers. We reported the association with smoking exposure and found that it was not consistently associated with our methylation scores across the cohorts (cohort specific association p-values of 5.4×10^{-5} , 3.4×10^{-5} , and 0.58, for CHILD, FAMILY, and START; original Table 3), possibly due to the low exposure in South Asian population (max weekly exposure was 42 hrs in contrast to 168 hrs in FAMILY and 98 hrs in CHILD). Meanwhile, air pollution data are currently not available. Here we additionally performed the association between maternal smoking and air pollution methylation score, using key CpGs from the largest air pollution EWAS to-date (Gondalia et al., 2021). However, there was no association between the air pollution score and any maternal smoking phenotypes ($p > 0.4$).

(4) One of the European cohorts and half of the South Asian cohort had DNA methylation measured on only 2500 CpG sites. This set of sites included only 125 sites previously linked to prenatal smoking. The resulting model of prenatal smoking was small (only 11 CpG sites). It is possible that a large model may have been more powerful.

That is correct – also see our response to R2 comment #1. In our previous analysis, we validated two scores (one based on CpGs on the < 3,000 CpGs array and the other one for the full HM450K). The score with more CpGs indeed had slightly better performance. We included this as one of the limitations of the paper. Nevertheless, it does not impact the conclusion that the scores (based on a larger or smaller model) are transferrable to diverse populations and can be used to comparatively study the DNAm influence of maternal smoking in newborns.

The following was added in the discussion:

“First, the customized array with a limited number of CpGs (<3,000) was designed in 2016 and many large EWASs on smoking and maternal smoking conducted more recently had not been included.”

(5) The health outcomes investigated are potentially interesting but there are other possibly more important outcomes of interest such as birth complications, asthma, and intellectual impairment which are known to be associated with prenatal smoking.

We thank the reviewer for bring up this point. One of the key health outcomes in the CHILD study was asthma, and data at later time points are available. However, we do not have similar outcomes collected in the other two studies (FAMILY and START), which focused on cardiometabolic health in young children. Thus, we did not initially include outcomes that were not available across all cohorts as the intention was to contrast the effects between populations.

We recognize that this is an important question and decided to provide the association results for asthma and allergy at available time points in CHILD, FAMILY, and START. We also included mode of delivery via emergency C-section as an additional proxy outcome of birth complications. However, none of these were marginally ($p < 0.05$) associated with the DNAm smoking score. These are now included in the updated Supplementary Table 8.

Reviewer #1 (Recommendations For The Authors):

(1) The number of samples in the South Asian birth cohort given in the abstract ($n = 887$) does not match the sample size of the START cohort from the results section (results, page 7, line 139, $n = 880$). It is also different from the final analytical dataset size from the methods section (page 17, line 386, $n = 890$). Please clarify.

We thank the reviewer for pointing this out. In the abstract, it was the final sample sized used for EWAS (no missingness in smoking history). The 880 in result was a typo for 890, which contains three individuals with missing smoking data. These have been updated with the correct sample size for START cohort that had full epigenome-wide methylation data ($n = 504$, and 503 with non-missing smoking history).

(2) Page 3, line 54: "consistent signal from the GFI1 gene ($ps < 5 \times 10^{-5}$)". Is ps a typo? If not then it might be clearer to state how many sites this included.

No, these summarized the six CpG sites in the GFI1 gene as outlined in Table 2. We have clarified in the abstract to show the number of CpG sites included.

(3) Please report effect sizes together with information about the statistical significance (p values).

We have updated the manuscript with (standardized) effect sizes whenever possible along with p -values.

(4) Page 4, line 80. This paragraph could be improved by adding a sentence explaining DNA methylation.

We thank the reviewer for this suggestion. A sentence was included to introduce DNAm at the beginning of the second paragraph:

"DNA methylation is one of the most commonly studied epigenetic mechanisms by which cells regulate gene expression, and is increasingly recognized for its potential as a biomarker (13)."

(5) Page 4, line 84. Sentence difficult to understand, please rephrase: "Our recent systematic review of 17 cord blood epigenome-wide association studies (EWAS) demonstrated that out of the 290 CpG sites reported, 19 sites were identified in more than one study; all of them associated with maternal smoking".

We have revised to clarify the review was on cord blood EWAS with five outcomes: maternal diabetes, pre-pregnancy body mass index, diet during pregnancy, smoking, and gestational

age.

“Our recent systematic review of 17 cord blood epigenome-wide association studies (EWAS) found that out of the 290 CpG sites reported to be associated with at least one of the following: maternal diabetes, pre-pregnancy body mass index (BMI), diet during pregnancy, smoking, and gestational age, 19 sites were identified in more than one study and all of them associated with maternal smoking.”

(6) Page 5, line 93. The second part of the sentence is not necessary: “The majority of cohort studies have focused on participants of European ancestry, but few were designed to assess the influence of maternal exposures on DNA methylation changes in non-Europeans”.

We have revised accordingly to:

“Only a handful of cohort studies were designed to assess the influence of maternal exposures on DNA methylation changes in non-Europeans.”

(7) Page 5, line 95. “It has been suggested that ancestral background could influence both systematic patterns of methylation (27), such as cell composition and smoking behaviours (28)”. The sentence is slightly unclear. Could it be rephrased to say that cell composition differences may be present by ancestry, which can lead to differential DNAm patterns?

We have revised accordingly to:

“It has been suggested that systematic patterns of methylation (Elliott et al., 2022), such as cell composition, could differ between individuals of different ancestral backgrounds, which could in turn confound the association between differential DNAm and smoking behaviours (Choquet et al., 2021).”

(8) Page 5, line 108. How does reducing the number of predictors lead to more interpretable effect sizes?

This was meant as a general comment in the context of variable selection, whereby the fewer predictors there are, the effect size of each predictor becomes more interpretable. However, we recognize this comment might be irrelevant to the specific approaches we adopted. We have revised it to motivate methylation score as a powerful instrument for analysis:

“Reducing the number of predictors and measurement noise in the data can lead to better statistical power and a more parsimonious instrument for subsequent analyses.”

(9) Page 5, line 112. Health consequences seem a bit strong, given that the analysis describes correlations/associations.

We have revised it to “association with”:

“In this paper, we investigated the epigenetic signature of maternal smoking on cord blood DNA methylation in newborns, as well as its influence on newborn and later life outcomes in one South Asian which refers to people who originate from the Indian subcontinent, and two predominantly European-origin birth cohorts.”

Results

(10) It would be very helpful to have a flow diagram to detail all of your analyses.

We thank the reviewer for this suggestion. In the revised manuscript, we have provided a flow chart as the new Figure 1, updated the summary of analysis in . Table 3, and added a new Supplementary Table 5 for the DNAm score derivation, as well as more detailed description of the statistical analysis in the Materials and Methods under the subsection “Statistical analysis”.

(11) Page 7, line 138. Please add a reference to the CHILD study.

We have added a reference of the CHILD study.

(12) Tables in results and in supplemental data a) contain a mixture of fields describing the newborn and its mother (this is not true for Supplementary Table 2), b) lack column descriptions, c) lack descriptions of abbreviations and formatting used in tables, d) use different font types, e) lack descriptions of statistical tests that were used to obtain p-values, f) use inconsistent rounding. Please correct and add the missing information.

We have consolidated the notation and nomenclature in all Tables and text. All numerical results are now rounded to 2 decimal places. The tests used were included in the Table headers as well as described in the Materials and Methods:

“For continuous phenotypes, an analysis of variance (ANOVA) using the F-statistics or a two-sample t-test was used to compare the mean difference across the three cohorts or two groups, respectively. For categorical phenotypes, a chi-square test of independence was used to compare the difference in frequencies of observed categories. Note that three of the categories under smoking history in the START cohort had expected cell counts less than 5, and was thus excluded from the comparison, the reported p-value was for CHILD and FAMILY.”

(13) Table 1. Sample sizes given in column descriptions do not add up to 1,650 (legend text).

We thank the reviewer for pointing this out. The updated sample size is 1,267, based on the 352 CHILD samples, 411 FAMILY samples, and 352 START samples. Notice that we did not remove those without full smoking history data as Table 1 was intended for the epigenetic subsamples.

(14) Page 7, line 156. Supplementary Tables are incorrectly numbered. In the text, Supplementary Table 4 comes after Supplementary Table 2.

We thank the reviewer for catching this and have corrected the ordering of the Supplementary Tables and Figures.

(15) Page 7, line 158. “cell compositions” - do you mean estimated white cell proportions?

We have revised it to “estimated cord blood cell proportions” in the text throughout.

(16) Smoking EWAS - do you see any overlap/directional consistency with the top findings from adult EWASs of smoking such as AHRR?

We annotated the top EWAS signals from the literature in the meta-analysis (new Figure 2; Supplementary Figures 1 and 3), but was only able to confirm associations in the GFI1 gene. The AHRR signals were also annotated, but below the FDR correction threshold as seen in new Figure 2 at the start of chromosome 5. We further added a new Supplementary Figure 3 to show the directional consistency with top findings (2,620 CpGs reported and 128 CpGs

overlapped with our meta-analysis) from Joubert et al., 2016. The Pearson's correlation coefficient with meta-analyzed effect for maternal smoking was 0.72 and for smoking exposure was 0.60.

We added the following to Results:

“Further, we observed consistency in the direction of association for the 128 CpGs that overlapped between our meta-analysis and the 2,620 CpGs with evidence of association for maternal smoking (19) (Supplementary Figure 3). Specifically, the Pearson's correlation coefficient for maternal smoking and weekly smoking exposure was 0.72 and 0.60, respectively.”

(17) Page 8, line 169. “also coincided with the *GFI1* gene” this is a bit imprecise. Please report the correlation with the CpG from the maternal smoking analysis.

The CpG was inside the *GFI1* gene, we have included the Pearson's correlation with the top hit in the text below:

“There were no CpGs associated with the ever-smoker status at an FDR of 0.05, though the top signal (cg09935388) was also mapped to the *GFI1* gene (Pearson's r^2 correlation with cg12876356 = 0.75 and 0.68 in CHILD and FAMILY, respectively; Supplementary Figure 1).”

(18) Page 8, line 171. Typo “ccg”: “cpg01798813”.

It has been corrected to “cpg01798813”.

(19) Page 8, line 176. Please be clear about the phenotype used in these analyses.

The EWAS of weekly smoking exposure in START was removed in this version of the manuscript, in reflection of the results and the reviewer's comments, as a result of this phenotyping being skewed and possibly leading to only spurious results (also see response to comment #20).

We have clarified the phenotypes for these results under “Epigenetic Association of Maternal Smoking in White Europeans” below:

“The maternal smoking and smoking exposure EWASs in CHILD did not yield any CpGs after FDR correction (Supplementary Figure 3).”

(20) What was the genomic inflation for the EWASs? 474 loci in the South Asian EWAS seems like a lot of findings. Perhaps a more robust method (e.g., OSCA MOMENT) might help to control the false positive rate.

The genomic inflation factor was moderately across the cohorts for smoking exposure: 1.02 in CHILD, 0.94 in FAMILY, and 1.00 in START. However, there was more inflation in the tail of the distribution in START than the European cohorts. The empirical type I error rates at 0.01, 0.001, 0.00001, were high in START (x1.7, x5.7, and x165 times at each respective threshold), in contrast to CHILD (x1.06, x1.05, and x0.6) or FAMILY (x1.6, x1.9, and 0). The smoking exposure EWAS based on START was thus removed as these are likely false positives and there was very low smoking exposure to start with (11 reported weekly exposure between 2–42 hrs/week out of 462 with non-missing data). We have added the QQ-plots as well as the genomic inflation factor for the reported meta-analysis in the new Supplementary Figure 2. The following was added to the Results:

“There was no noticeable inflation of empirical type I error in the association p-values from the meta-analysis, with the median of the observed association test statistic roughly equal to

the expected median (Supplementary Figure 2)."

(21) *What is the targeted array? I don't think it has been introduced prior to this point.*

We introduced it in the Materials and Methods under subsection "Methylation data processing and quality controls". Considering this comment and previous comments on the ordering of Tables and Figures, we have decided to place Materials and Methods after Introduction and before Results.

(22) *The MRS section is described poorly in the results section. It is not clear where the 11 or 114 CpGs come from.*

We now include an analytical summary of all scores (derived or external from literature) in the new Supplementary Table 5. Further, we updated the description of scores in Materials and Methods under the subsection "Using DNA Methylation to Construct Predictive Models for Maternal Smoking" to clarify the source and types of MRSs derived:

"To evaluate whether the targeted GMEL-EPIC array design has comparable performance as the epigenome-wide array to evaluate the epigenetic signature of maternal smoking, a total of three MRSs were constructed, two using the 128 CpGs available in all cohorts – across the HM450K and targeted GMEL-EPIC arrays – and with either CHILD ($n = 347$ with non-missing smoking history) or FAMILY ($n = 397$) as the validation cohort, and another using 2,107 CpGs that were only available in CHILD and START samples with CHILD as the validation cohort. Henceforth, we referred to these derived maternal smoking scores as the FAMILY targeted MRS, CHILD targeted MRS, and the HM450K MRS, respectively."

(23) *Page 9, line 187. "There was no statistically significant difference between the two scores in all samples ($p = 1.00$) or among non-smokers ($p = 0.24$).". How was the significance assessed? Please describe the models (outcome, covariates, model type) used for comparing the two models. It would also be good to report the correlation between the scores.*

We have added a subsection "Statistical analysis" under Materials and Methods that described the tests. The correlation between scores is now summarized as a heatmap across all cohorts in the new Supplementary Figure 6.

"For each cohort, we contrasted the three versions of the derived scores using an analysis of variance analysis (ANOVA) along with pairwise comparisons using a two-sample t-test to examine how much information might be lost due to the exclusion of more than 10-fold CpGs at the validation stage. We also examined the correlation structure between all derived and external MRSs using a heatmap summarizing their pairwise Pearson's correlation coefficient."

(24) *Please include the number of samples in the training/validation and in the test set in the methods and in the results.*

We thank the reviewer for this suggestion. In the revised manuscript, we have provided a flow chart as the new Figure 1 and more detailed description of the method in the Materials and Methods. Please also see response to comment #22. The training sample size is based on Joubert et al., (2016), which is 5,647. For our main analyses, the validation sample with non-missing phenotypes remained the CHILD cohort ($n=347$), while the FAMILY ($n=397$) and START ($n=503$) samples were the independent testing data. We alternatively provided another scenario, in which the FAMILY sample was the validation cohort, while CHILD and START were the testing cohorts. The exact sample size and performance metrics for each scenario and score are clearly summarized in the new Supplementary Table 5.

(25) Table 3. Please clarify the type of information contained in the four last columns (p-value?).

Yes – these are the individual cohort p-values. We have taken the suggestion from comment #12 to fully describe all columns and fields.

(26) Page 10, line 215: "The meta-analysis revealed no heterogeneity in the direction nor the effect size of associations between populations". Please quote/refer to the results.

In the revision, the heterogeneity p-values were quoted and the relevant tables (Supplementary Table 8) were added to this sentence.

(27) Figure 2 has issues with x labels. Due to the low number of ever smokers in START, the boxplot may not be the best visualisation method. It would also benefit from listing n's per group.

We appreciate this comment to improve the figure presentation. We increased the font size for the X-labels. The sample size for each group in START was also labeled in the new Figure 3 (previously Figure 2).

Discussion

(28) Studying the association between maternal smoking and cord blood DNAm is interesting from a biological perspective as it allows for assessing the immediate and long-term effects of maternal smoking on newborn health. However, in terms of calculating the MRS, what are the benefits of using cord blood over the mother's blood? We know that blood-based DNAm smoking score is a powerful predictor of long-term smoking status.

The reviewer raises an interesting point – abundant literature supports that DNAm changes are tissue-specific. While mother's blood DNAm smoking score reflect the long-term exposure to smoking in mothers, the cord blood DNAm captures the consequence of such long-term exposure for newborn health. One of the key results of our study is showing that established DNAm signatures of maternal smoking, which is known to mediate birth size and weight in white Europeans (these references were cited in the original manuscript), carries the same effect of reducing birth weight and size in the South Asian population. This is a critical finding from a DoHaD and public health perspective, as DNAm signatures of maternal smoking, irrespective of the smoking status of the mother, can influence the health trajectory of the newborns.

We have expanded our discussion based on this suggestion to highlight the unique features of studying maternal smoking via different tissues and their implications. The following was added to the discussion:

"There are several advantages of using a cord blood based biomarker from the DoHaD perspective. Firstly, cord blood provides a direct reflection of the in utero environment and fetal exposure to maternal smoking. Additionally, since cord blood is collected at birth, it eliminates potential confounding factors such as postnatal exposures that may affect maternal blood samples. Furthermore, studying cord blood DNAm allows for the assessment of epigenetic changes specifically relevant to the newborn, offering valuable information on the potential long-term health implications."

(29) Page 13, line 285: "Fourth" without "third".

It has been revised accordingly.

Methods

(30) The methods section does not contain all the details required to replicate the analysis. Whenever statistical analysis is conducted, this section should clearly describe the type of the analysis (linear regression, t-test, etc.) and name the dependent and independent variables. Sample sizes should also be given.

We added further details of test used and sample size for each analysis. We have also included a new “Statistical analysis” subsection under Materials and Methods.

(31) Please describe MRS testing in the methods.

We tested MRS with respect to binary and continuous smoking phenotypes using a logistic and linear regression, respectively. The predictive value was assessed using area under the roc curve for the binary outcome and an adjusted R² for the continuous outcome. These were added to the new “Statistical analysis” subsection under Materials and Methods. See response to comments #22-24, and #30.

(32) Please describe the methods used to compare the two versions of MRS for maternal smoking.

It was a two-sample t-test, which was described in the Figure legends. We have now added this to the new “Statistical analysis” subsection under Materials and Methods.

(33) Please describe testing the associations between MRS and Offspring Anthropometrics in more detail.

We added further details on the regression model and the test for association in the methods. We have now added this to the new “Statistical analysis” subsection under Materials and Methods.

(34) Meta analysing the 450k and GMEL arrays is going to substantially reduce the number of CpGs under investigation.

We agree with the reviewer that this is not optimal for signal discovery. However, this is the only way we could synthesize evidence across the cohorts as FAMILY samples were only processed using the customized array. We added the following as a limitation of the study in the discussion.

“First, the customized array with a limited number of CpGs (<3,000) was designed in 2016 and many large EWASs on smoking and maternal smoking conducted more recently had not been included.”

(35) Page 16, line 364: GDM abbreviation was used in the results section (line 145), yet it is introduced in line 364.

Thank you for catching this, we have removed the duplicate.

(36) Page 17, line 381: Given the stated importance of ancestry, why not restrict the sample to genetically confirmed groups?

The reviewer has a valid point that ancestry, either perceived or genetic, can introduce additional heterogeneity due to potential differences in genetics, cultural and social practices, and lifestyles. Genetic data are indeed available for a subset of the individuals. In the original version of the manuscript, we used a stringent ancestry calling method by mapping all individuals with the 1000 Genomes samples from continental populations. The final definition was based on a combination of self-reported and genetically confirmed ancestry. However, if we restricted only to genetically confirmed groups, the sample size would be reduced to 312 (vs. 411), 268 (vs. 352), and 488 (vs. 504) in FAMILY, CHILD, and START, respectively.

We compared the mean difference in the beta-values of the top associated CpGs and the derived MRS between those genetically confirmed vs. self-reported ancestral groups, and observed no material difference. These results are now included in the Supplementary Materials as part of the sensitivity analysis. Thus, given these considerations, we decided to use this complementary approach to retain the maximum number of samples while ensuring some aspect of ancestral homogeneity.

“To maximize sample size in FAMILY and CHILD, we retained either self-identified or genetically confirmed Europeans based on available genetic data (Supplementary Table 1).”

| (37) Page 18, line 397: *sensitivity analysis not sensitive analysis.*

Thank you for catching this, we have revised accordingly.

| (38) Page 18, line 409: *smoking was rank transformed however, it would be good to see regression diagnostics for the lead loci in the EWAS to check that assumptions were met.*

We thank the reviewer for this suggestion. Smoking exposure is indeed skewed and in fact very much zero-inflated across the cohorts. The raw phenotype violated several model assumptions in terms of variance heteroskedasticity, outlying values (influential points), and linearity. The diagnostics suggested improved deviation from model assumption, yet some aspects of the violation remained at a lesser degree. We included a comparison of results before and after transformation and model diagnostics for the lead CpG using CHILD and FAMILY data in the Supplementary Materials. The following was added to the results:

“As a sensitivity analysis, we repeated the analysis for the continuous smoking exposure under rank transformation vs. raw phenotype for the associated CpG in GF11 and examined the regression diagnostics (Supplementary Material), and found that the model under rank-transformation deviated less from assumptions.”

| (39) Page 19, line 418: *FDR seems quite a lenient threshold, especially when genome-wide significance thresholds exist. I would be inclined to view the EWAS findings as null.*

The choice of use FDR to was indeed arbitrary as there has been no consensus on what significance threshold, if any, should be used in the context of EWAS. The significance threshold for GWAS (Pe'er et al., 2008) probably does not apply directly to EWAS as the number of effective tests will likely differ between genome-wide genetic variants and CpGs. The Bonferroni corrected p-value threshold in this context would be $0.05/200,050=2.510^{-7}$, which is still less stringent than the GWAS significance threshold. We originally decided to follow the convention of previous studies and use FDR to filter out a subset of plausible associations to contrast the top association signals for their effects between different populations and with reported effect sizes.

We have revised the manuscript throughout by removing the notion of significant associations, and instead used the phrase “top associated signals” or “top associations” when

discussion EWAS results for individual CpGs. The following was added to Materials and Methods to clarify the choice of our threshold:

“For each EWAS or meta-analysis, the false discovery rate (FDR) adjustment was used to control multiple testing and we considered CpGs that passed an FDR-adjusted p-value < 0.05 to be relevant for maternal smoking.”

(40) I do not understand Supplementary Figure 6 - how have the data been standardised? Why not plot the CpGs on the beta-value scale?

The standardized values were plotted as the reported p-values for the mean and variance equality tests (i.e. ANOVA F-test, Levene's test, Anderson-Darling test) were based on these transformed values to reduce inflation due to non-normality. We have since removed this comparison and kept only the comparison of the overall score as the number of CpGs in the HM450k score (143 CpGs) for comparison is too high to be visually interpretable.

(41) It is my understanding, that the MRS for maternal smoking was constructed using external weights projected and regularised using elastic net (effectively trained) in CHILD cohort. The results section discusses associations between maternal smoking history and outcomes in CHILD, FAMILY, and START. Training and testing the score in the same sample (cohort) may result in overfitting and therefore should not be implemented.

The original MRS was constructed using external weights from an independent discovery sample (Joubert et al., 2016; $n > 5,000$) and the LASSO validation was done in CHILD ($n = 352$), external testing was in FAMILY and START. This was the lassosum framework whereby we leverage larger sample size from external studies to select more plausible CpGs as candidates to include in the model. Thus, training, validation, and testing were not done in the same samples. We have included a Figure 1 to illustrate the updated analytical flow and a graphical abstract to summarize the methods.

(42) Is it a concern that the findings don't seem to replicate Joubert's results, which came from a much larger study?

Replication is usually done in samples much larger than the discovery samples, thus it is not a concern that we were unable to confirm all signals from Joubert et al., (2016). However, 6/7 of the top associations (FDR adjusted p-value < 0.05) in the meta-analysis were declared as significant in Joubert et al. (2016). In addition, the fact that using Joubert's summary statistics, we were able to derive MRSs that were strongly associated with both smoking history and weekly exposure suggests shared signals. Also see response to R1 comment #16 for a comparison of effect consistency.

(43) Please check that all analysis scripts have been uploaded to Github and that the EWAS results are publicly available.

We thank the reviewer for this suggestion. All updated scripts and EWAS results are available on Github. We are working to have the results also submitted to EWAS catalog.

Reviewer #2 (Recommendations For The Authors):

The impact of this study is reduced due to previous findings:

(1) Previous studies have already shown that DNA methylation may mediate the effect of maternal smoking on birth size/weight (see e.g. <https://doi.org/10.1098/rstb.2018.0120>; <https://doi.org/10.1093/ije/dyv048>).

We thank the reviewer for this point and would like to take the opportunity to clarify that it was not our objective to examine whether there was a causal relationship, between DNA methylation and birth size that was mediated by maternal smoking. One of the key messages of our study is to evaluate whether epigenetic associations – at individual CpGs and aggregated as a score – are consistent between white European and South Asian populations. One way to examine this is through using established DNAm signatures of maternal smoking, which is known to mediate birth size and weight in white Europeans (these references were cited in the original manuscript), and confirm whether they also carry the same effect on birth outcomes in the South Asian population.

Indeed, our results support that maternal smoking methylation score was consistently associated with negative outcomes in newborns of both white European and South Asian mothers despite no maternal smoking was present in South Asian mothers. These collective point to the possibility that the maternal smoking MRS was capturing a lot more than just smoking and second-hand smoking, but potentially other environmental exposures that also lead to oxidative stress. These together are associated with health consequences, including reduced birth size/weight. One of the candidates for such exposure is air pollution as some of the maternal smoking CpGs were previously linked to air pollution. However, we were unable to assess this hypothesis directly without the air pollution data, and the air pollution methylation score was not associated with smoking history (Supplementary Figure 5) nor smoking exposure ($p > 0.4$ in CHILD, FAMILY and START).

The following was added to Materials and Methods under the subsection Using DNA Methylation to Construct Predictive Models for Maternal Smoking:

“To benchmark and compare with existing maternal smoking MRSs, we calculated the Reese score using 28 CpGs (48,49), Richmond score using 568 CpGs (49), Rauschert score using 204 CpGs (50), Joubert score using all 2,620 CpGs with evidence of association for maternal smoking (19), and finally a three-CpG score for air pollution (51). The details of these scores and score weight can be found in Supplementary Table 4.”

The following was added to Results

“Both produced methylation scores that were significantly associated with maternal smoking history (ANOVA F-test p -values = 1.0×10^{-6} and 2.4×10^{-14} in CHILD and 6.9×10^{-16} and $< 2.2 \times 10^{-16}$ in FAMILY), and the best among alternative scores for CHILD and FAMILY (Supplementary Table 5). With the exception of the air pollution MRS, all remaining scores were marginally associated with smoking history in both CHILD and FAMILY (Supplementary Figure 5).”

(2) Due to the small study size and low levels of prenatal smoke exposure, the model derived here is of little value and is, in fact, superseded by a previously published model (PMID: 27323799). At the very least, the model should be evaluated here. A novel aspect of this study is the inclusion of a South Asian cohort. Unfortunately, smoke exposure is practically non-existent, so it is unclear how it can be used. The more interesting finding in this study is the possibility that environmental factors such as second-hand smoke or pollution may have similar effects on pregnancies as maternal smoking. Are these available? If so, they could be evaluated for associations with DNA methylation. This would be novel.

In the revised manuscript, we included the Reese score (Reese et al., 2017) and a few other maternal smoking scores for comparison. In the CHILD cohort, the performance was comparable to our derived score (AUC of 0.95 vs. 0.94 for Reese score), but its applicability was limited since the FAMILY dataset was profiled using a targeted array and only 7 out of 28 of the CpGs in the Reese score were available (AUC of 0.89 vs. 0.72 for Reese). As compared to

the remaining scores from literature (see the new Supplementary Table 5 for complete results), Reese's score has generally favorable performance.

We did examine second-hand smoking in the original manuscript, showing a significant association with weekly maternal smoking exposure (original Table 3 and Supplementary Table 8). However, air pollution data is not available for assessment.

(3) The other novel aspect is the evaluation of associations with outcomes later in life. Height and weight are interesting but impact could be gained by including other relevant outcomes such as birth complications, asthma, and intellectual impairment which are known to be associated with prenatal smoking.

We thank the reviewer for bring up this point. One of the key health outcomes in the CHILD study was asthma, and data at later time points are available. However, we do not have similar outcomes collected in the other two studies (FAMILY and START), which focused on cardiometabolic health in young children. Thus, we did not initially include outcomes that were not available across all cohorts as the intention was to contrast the effects between populations.

We recognize that this is an important question and decided to provide the association results for mother reported asthma and allergy, but based on different definitions as these outcomes cannot be harmonized across the cohorts. We also included mode of delivery via emergency C-section as an additional proxy outcome of birth complication.

The following was added to Materials and Methods:

“Mode of delivery (emergency c-section vs. other) was collected at the time of delivery.”

“Additional phenotypes included smoking exposures (hours per week) at home, potential allergy based on mother reporting any of: eczema, hay fever, wheeze, asthma, food allergy (egg, cow milk, soy, other) for her child in FAMILY and START, and asthma based on mother's opinion in CHILD (“In your opinion, does the child have any of the following? Asthma”).”

The following was added to Results:

“The maternal smoking MRS was consistently associated with increasing weekly smoking exposure in children reported by mothers at the 1-year (0.51 ± 0.15 , FDR adjusted $p = 0.0052$), 3-year (0.53 ± 0.16 , FDR adjusted $p = 0.0052$), and 5-year (0.40 ± 0.15 , FDR adjusted $p = 0.021$) visits with similar effects.”

“We did not find any association with self-reported allergy or asthma in children at later visits (Supplementary Table 8). Further, there was no evidence of association between the MRS and any maternal outcomes (Supplementary Table 8).”

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